

CHUNYANG LOU

Geospatial Software Engineer | GIS, Remote Sensing & 3D Geological Modeling

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SUMMARY

Geospatial Software Engineer with 5 years of applied professional experience combining 3D geological modeling, GIS, remote sensing, AI-assisted geospatial analysis, and WebGIS platform development. Experienced with authoritative geoscience data workflows, including GOCAD/SKUA-style geological objects, drillhole and subsurface datasets, lithological domains, terrain models, remote-sensing imagery, PostGIS-backed spatial databases, and interactive 2D/3D web visualization. Strong ability to translate geological concepts into reproducible software systems, with emphasis on data provenance, QA/QC, coordinate integrity, section generation, and stakeholder-facing technical products.

TECHNICAL SKILLS

3D Geological Modeling	SKUA-GOCAD, Leapfrog Geo, Petrel, Move, implicit surfaces, lithological domains, fault and stratigraphy surfaces, cross-sections, fence diagrams, voxel/block model concepts
GIS & Spatial Analysis	ArcGIS Pro, ArcMap, QGIS, ArcGIS Online, cartography, georeferencing, coordinate systems, spatial joins, overlay analysis, hydrological and catchment analysis, topology QA/QC
Remote Sensing & Terrain	ENVI, ERDAS Imagine, satellite imagery, aerial imagery, LiDAR, DEM/DSM, hillshade, slope, DEM-of-Difference, land-cover and disturbance classification, change detection
Full Stack / WebGIS	React, TypeScript, JavaScript, Python, REST APIs, Node.js/FastAPI-style services, Mapbox, Leaflet, deck.gl concepts, vtk.js/WebGL, Vite, Vercel, interactive dashboards
Databases & Cloud	PostgreSQL/PostGIS, SQL Server, Oracle Spatial concepts, ETL, schema design, metadata, spatial indexing, object storage/CDN concepts, reproducible data packages
AI & Data Science	Machine-learning fundamentals, AI-assisted geospatial QA, feature extraction concepts, Python data science stack, NumPy, Pandas, GeoPandas, rasterio, statistical spatial analysis, geostatistics, kriging, variography

PROFESSIONAL EXPERIENCE

NV5 Geospatial — Lakewood, CO

Aug 2025 – Mar 2026

Full Stack Software Engineer & WebGIS Developer — Geospatial, Terrain and Remote-Sensing Platforms

- Built and maintained WebGIS and geospatial analysis workflows supporting LiDAR terrain analysis, river-corridor mapping, remote sensing, and natural-resource projects.
- Developed reproducible Python and JavaScript/TypeScript workflows for DEM processing, imagery preparation, spatial data QA/QC, and geospatial product delivery.
- Designed interactive web maps and dashboards using ArcGIS Online, Mapbox/Leaflet-style tools, PostGIS-backed datasets, and client-facing visualization patterns.
- Integrated LiDAR point clouds, DEMs, aerial imagery, satellite imagery, GPS and field-survey datasets into clean spatial products with controlled coordinate-reference systems.
- Implemented automation for batch processing, data validation, map production, metadata generation, and repeatable geoprocessing across large multitemporal datasets.
- Supported AI-assisted and rules-based feature extraction concepts for land-cover, disturbance, channel morphology, and terrain-change interpretation.
- Collaborated with engineers, surveyors, environmental scientists, and project managers to deliver technically defensible spatial products for stakeholders.

Geologist & 3D Geological Modeler — Subsurface, Sediment and Mining Systems

- Built 3D geological models using Leapfrog Geo, SKUA-GOCAD and Move, producing implicit surfaces, lithological domains, sediment-body models and subsurface interpretations.
- Integrated drillhole, core, geochemical, geophysical, survey and remote-sensing datasets into unified spatial databases for multidisciplinary geological interpretation.
- Generated cross-sections, fence diagrams, 3D visualizations, and surface/solid models to communicate geological structure, sediment bodies, valley-fill stratigraphy and mining-related materials.
- Applied geostatistical methods such as variography and kriging to interpolate geological and material properties and define spatial domains for interpretation.
- Performed rigorous QA/QC on drillhole collars, survey data, lithology intervals, assay tables, model geometry, coordinate systems and source provenance before delivery.
- Developed Python workflows to clean, transform, validate and document large geological datasets, reducing manual effort and improving reproducibility.
- Worked with geologists, engineers and environmental scientists to translate geological concepts into defensible digital models and client-ready products.

SRK Consulting — Vancouver, BC**Feb 2022 – Jan 2023*****Geologist / GIS Analyst — Mining, Environmental and Catchment Analysis***

- Supported mining, environmental and water-related consulting projects with geological interpretation, GIS analysis, remote-sensing preparation and spatial data management.
- Mapped lithology, sediment, tailings, disturbed ground, catchments, waterways and floodplain areas to support environmental assessment of mining-affected landscapes.
- Digitized and georeferenced historical maps and aerial photographs to reconstruct past channel positions, mining disturbance and land-use change over time.
- Prepared thematic maps, spatial datasets, report figures and technical deliverables for multidisciplinary project teams and regulatory documentation.
- Wrote Python scripts for batch map exports, attribute updates, spatial data checks and repetitive geoprocessing tasks.
- Maintained spatial databases with consistent naming, metadata, projection conventions and QA/QC checks for geometry, attributes and coordinate reference systems.

SELECTED PROJECT EXPERIENCE

- 3D Geological WebViewer and Section-Ready Data Package: contributed to a workflow concept that separates backend geological computation from browser display, emphasizing valid lithology policy, fault/stratigraphy overlays, section metadata, binary chunk layouts and fail-closed visualization contracts.
- Remote-Sensing Change Detection: applied multitemporal imagery, DEMs and LiDAR products to identify terrain change, disturbance patterns, channel migration and environmental impacts in mining and river-corridor contexts.
- Subsurface Data Integration: combined drillhole, core, geological, geochemical, geophysical and survey datasets into coherent 3D models with explicit provenance and QA/QC standards.
- WebGIS Decision Support: built interactive maps, dashboards and geospatial data packages for technical teams and non-technical stakeholders, focusing on clarity, performance and reproducibility.

EDUCATION

Bachelor of Science in Geosciences, Geology Emphasis**2014-2018**

University of Arizona — Tucson, AZ

Bachelor of Science in Computer Science**2018-2021**

Columbia University, Columbia Engineering — New York, NY